

- 5 (a) The resistance  $R$  of a metal is given by

$$R = \rho \frac{L}{A}$$

where  $\rho$  is the resistivity of the material of a wire,  $L$  is the length of the wire and  $A$  is the circular cross-sectional area of the wire.

An experiment is performed to determine the resistivity  $\rho$  of the wire. The average of the measurements, with their actual uncertainties, are :

| diameter of wire / mm | length of wire, $L$ / cm | resistance of wire, $R$ / $\Omega$ |
|-----------------------|--------------------------|------------------------------------|
| $0.23 \pm 0.01$       | $40.0 \pm 0.1$           | $4.80 \pm 0.05$                    |

Calculate the actual uncertainty in  $\rho$ .

actual uncertainty in  $\rho = \dots\dots\dots \Omega \text{ m}$  [3]

- 5 (b) A battery of electromotive force (e.m.f.) 6.0 V and internal resistance  $r$  is connected to an external resistor of resistance  $12\ \Omega$  and a NTC thermistor X, as shown in Fig. 5.1.

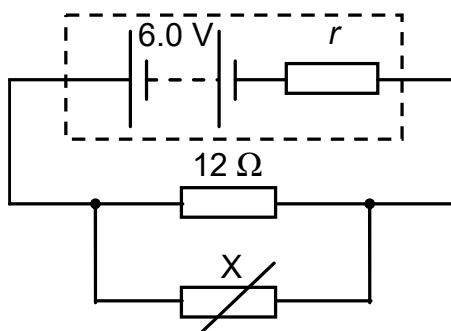


Fig. 5.1

- (i) Explain, in terms of energy changes in the circuit, why the potential difference across the terminals of the battery is less than the e.m.f.

.....  
 .....  
 ..... [2]

- (ii) The combined resistance of the  $12\ \Omega$  resistor and thermistor X connected in parallel is  $4.8\ \Omega$ .

Determine the ratio

$$\frac{\text{power dissipated in thermistor X}}{\text{power dissipated in } 12\ \Omega \text{ resistor}}.$$

ratio = ..... [3]

- (iii) The temperature of NTC thermistor X is now decreased.

State and explain the effect, if any, of this temperature change on the total power produced by the battery.

.....  
 .....  
 ..... [2]

- (c) The  $I$ - $V$  characteristic of a 60 W filament lamp is shown in the fig. 5.2.

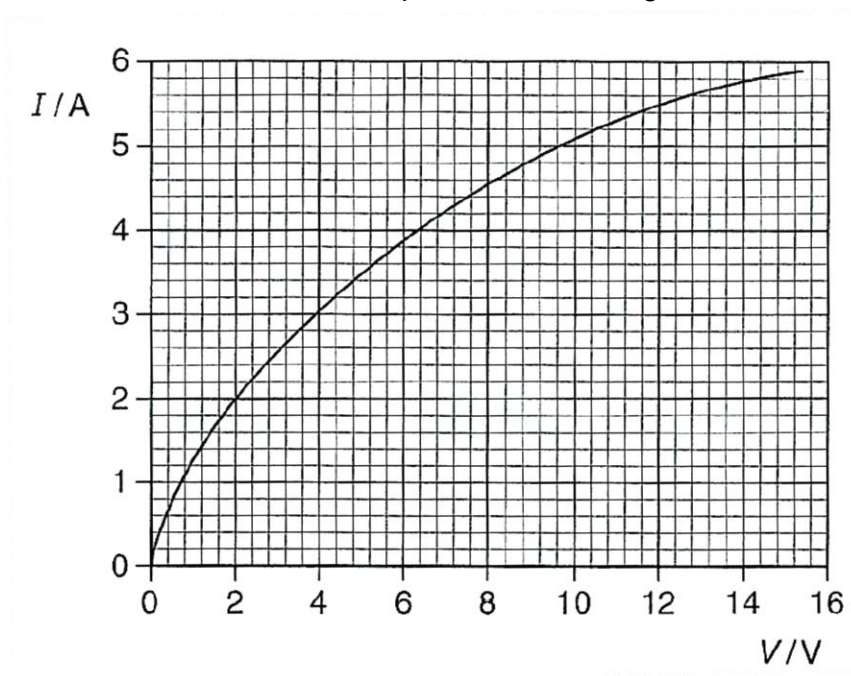


Fig. 5.2

- (i) Suggest why the graph stops at 15.4 V.

.....  
 ..... [1]

- (ii) Explain why the resistance of the lamp is **not** the reciprocal of the gradient of the graph.

.....  
 .....  
 ..... [1]

[Total: 12]